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COVER SHEET

Executive Summary, DARPA Defense Sciences Office (DSO)
Office-wide Broad Agency Announcement HR001126S0015

Executive Summary Title	Choosing the Next Experiment: Finding Printable Alloys at One Tenth the Cost
DSO Thrust Area	(1) Materials, Manufacturing, and Energy Systems; (4) Complex, Dynamic, and Intelligent Systems
Proposer Organization	Brigham Young University (BYU), Provo, Utah

Technical Point of Contact (POC)

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Executive Summary Technical Description

What is the proposed work attempting to accomplish or do? We aim to cut the cost and time of finding and qualifying a new printable alloy by about ten times, and to do the same for other multi-step materials workflows. Developing an alloy for laser powder bed fusion (LPBF), a form of metal 3D printing, takes five steps: a thermodynamic calculation (minutes), powder production (a day), printed tracks and cubes (hours), electron microscopy (a day), and tensile tests (days). We will build and prove a method that decides, as each result arrives, which step to run next and on which candidate alloy.

How is the work performed today, and what are the limitations? Experimental campaigns usually send every candidate through every step, so much of the budget goes to full tests of alloys that a cheap early measurement could have ruled out. Computational campaigns work as a funnel: many simulations, then a few survivors go to experiments. The funnel is only as good as its physics. Standard thermodynamic screens miss effects such as zirconium additions that keep printed high-strength aluminum from cracking (Martin et al., *Nature* 2017), so the funnel can discard the best alloys. Published methods that learn from partial results optimize a single property, wait for each result before choosing the next experiment, and have been tested only on synthetic problems.

What is new in our approach, and why will it succeed? Each step gets its own statistical model, and the models are linked in the order of the steps, so uncertainty carries through the chain (Astudillo and Frazier, NeurIPS 2021). For each possible experiment, the algorithm divides the expected gain by the cost and runs the best one, so a candidate can stop at any step (Buathong et al., ICML 2024). Simulations run continuously at full throughput on BYU's computing cluster, and each choice uses all data available at that moment. We will add three things not done before: balancing several properties at once (freedom from cracks, strength, ductility); choosing new experiments while others are still running; and a demonstration on a real five-step workflow with build-to-build scatter. In a hand-built simulation of this chain whose crack screen ignores zirconium, the screening funnel stalled at 63% of the best possible alloy after $100,000. The algorithm reached that level after about $26,500, or 3.8 times less spend (90% confidence interval 2.0 to 5.7; 20 campaigns). When the screen's physics was right, screening kept pace. The simulation is noise-free, with one objective and one experiment at a time, so ten times is our target, not yet a result. The full chain already runs for aluminum alloys: our ultrasonic atomizer makes powder of four compositions per week, LPBF builds cost $800 each at a University of Utah facility, and the BYU Microscopy Center supplies electron microscopy. We have completed a powder-to-part demonstration on a baseline aluminum alloy. The PI developed Honegumi, open software that helps experimentalists set up such searches, and co-authored the *Chemical Reviews* survey of self-driving laboratories.

Who will care, and what will the impact be if the work is successful? Qualifying a new material for a Department of War (DoW) system takes many expensive tests in a fixed order. A method that picks the next test and states its confidence shortens that path for any multi-step materials or manufacturing workflow. The first target is high-strength aluminum alloys that print without cracking, for aircraft structures and for replacement parts no longer made. The method moves to a new workflow by swapping in its steps and their costs. A DARPA-scale effort would add a plasma melting module for our atomizer (about $150,000), extending the chain to refractory alloys for hypersonic structures, and more printer capacity to run many builds at once. We will release software, data, and models openly for DoW laboratories to use on their own materials.